From Engagement Metrics to Precision Care:

Computational Phenotyping for AI-Driven Digital Mental Health

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Background

Digital mental health interventions (DMHIs) have expanded rapidly, yet most platforms evaluate success through aggregate engagement metrics such as logins and completion rates[5], assuming a linear relationship between engagement volume and clinical improvement. Emerging evidence challenges this assumption: engagement alone does not consistently reduce distress[2]. Highly engaged users may still experience elevated burnout, while less engaged users may possess protective coping mechanisms unrelated to technology.

Current DMHI frameworks fail to account for the complex, non-linear interactions between technological interaction and psychological burden.

The "digital divide" compounds this problem: age-related differences in digital literacy and technostress may cause the same intervention to benefit younger users while inadvertently increasing burden for older adults[1]. A critical gap exists in identifying distinct behavioral phenotypes that capture how age, technology engagement, and distress interact to shape user needs. Moving beyond one-size-fits-all analytics toward computational phenotyping[3,4] can enable AI-driven, precision-tailored digital mental health care.

Methodology

This pilot study utilized unsupervised machine learning to identify distinct behavioral phenotypes, moving beyond aggregate averages to uncover how users heterogeneously interact with digital tools under varying levels of distress.

Sample

•N = 181 adults enrolled in a digital behavioral toolkit trial

Measures

•Technology engagement: Technology Engagement: Calculated as the mean of 7 standardized items assessing active usage and platform interaction (tech_1–tech_7)

•Perceived distress: Calculated as the mean of 6 standardized items measuring current psychological burden (prodis1–prodis6)

•Age: Treated as a continuous variable to capture generational shifts in digital literacy.

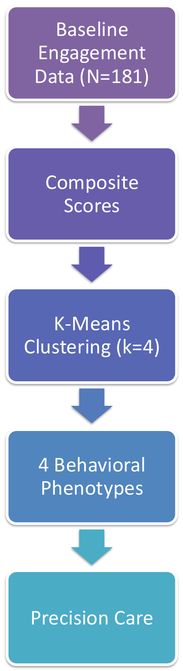
Analysis

•Z-scored variables → K-means (k = 4, nstart = 100)

•Elbow & silhouette: 4-cluster solution (avg sil. = 0.30)

•PCA visualization (Dim1 42.7%, Dim2 31.1%)

Analytic Pipeline



Age + Tech Engagement + Distress → Tailored AI Intervention Pathways
Z-scored • Silhouette validated (0.30) • PCA projected

Results

K-means clustering revealed four distinct behavioral phenotypes (each representing 23–27% of the sample) based on age, technology engagement, and psychological distress. These findings demonstrate that high technology engagement does not inherently buffer against psychological burden, as evidenced by the divergent distress levels observed in high-engagement groups.

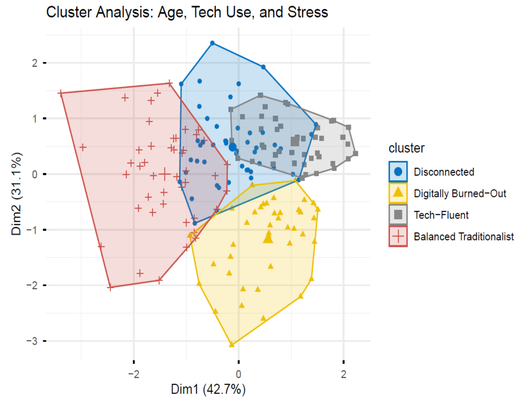


Fig 1. K-means cluster solution in PCA space (Dim1 42.7%, Dim2 31.1%). Four balanced phenotypes (n = 42–49 per cluster).

Key Findings:

•**The High-Engagement Paradox:** Highly engaged users appeared in both the lowest and highest distress clusters, suggesting that engagement volume alone is a poor proxy for mental health status.

• **Significant Heterogeneity:** All clusters differed significantly across age, engagement levels, and distress scores (p<0.001)

•**Identifying "Burnout":** The Digitally Burned-Out phenotype characterized younger users with the highest engagement (4.35) and highest distress (3.29), highlighting a high-risk group invisible to traditional aggregate metrics

Results

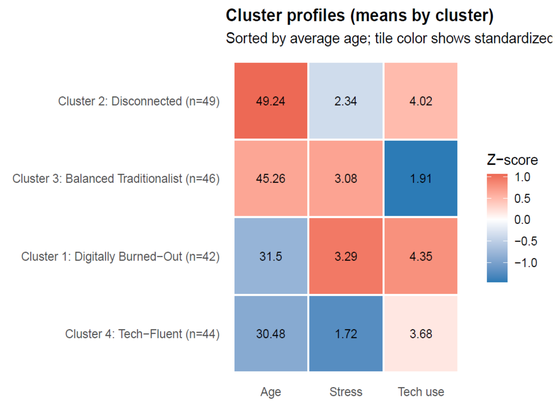


Fig 2. Cluster profile means (Z-scores). Clusters differ on Age, Engagement, and Distress (all p < .001).

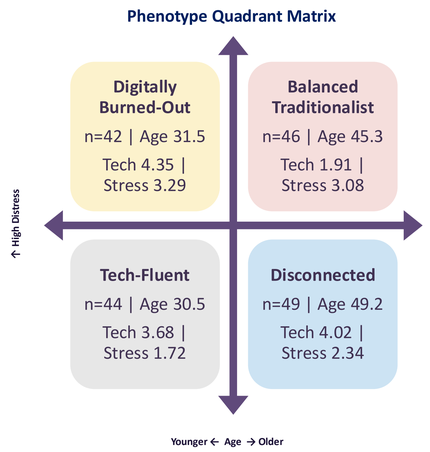


Fig. 3 Phenotype Quadrant — age × distress mapping of four K-means clusters (N = 181)

Clusters differed significantly on age, engagement, and distress (all p < .001).
Correlation: age–tech r = –0.22; age–stress r = 0.09; tech–stress r = –0.11

Final Statistical analysis/tests/ text

Conclusion

1. **Engagement ≠ protection.** High tech use appeared in both high- and low-distress clusters; volume alone does not buffer distress.
2. **Phenotypic heterogeneity matters.** Four balanced clusters (23–27% each) with distinct profiles were invisible to aggregate metrics (p < .001).
3. **One-size-fits-all widens gaps.** Uniform digital approaches risk excluding older low-tech users and overlooking screen-driven distress.

Implications

- **Adaptive onboarding:** Route Burned-Out users to boundary-setting modules; Traditionalists to simplified interfaces.
- **Precision engagement:** Tailor content intensity, modality, and pacing per phenotype rather than maximizing uniform engagement.
- **Equity-centered policy:** Mandate phenotype-level reporting so digital tools serve diverse age & literacy profiles equitably.

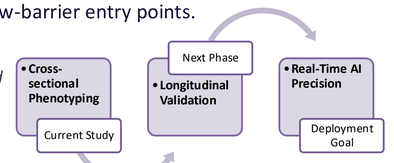
Future Directions: Scaling AI Precision

This pilot establishes computational phenotyping as a scalable bridge from engagement analytics to AI-driven precision care. Three priorities guide the next phase:

- **Longitudinal validation:** Track phenotype stability across T1→T2 using PHQ-9 and GAD-7 clinical change scores; integrate emotion regulation (DERS) and self-efficacy as mediators.
- **Predictive risk-flagging:** Build models that detect early signs of disengagement or distress escalation, shifting care from reactive to anticipatory.
- **Adaptive deployment:** Pilot phenotype-informed onboarding in a live digital toolkit — Burned-Out users receive boundary-setting prompts at intake; Traditionalists get simplified, low-barrier entry points.

Research Trajectory:

Goal: *Move from "did they log in?" to "did the right person get the right support at the right time?"*



Phenotype-to-Intervention Mapping

Digitally Burned-Out → Digital boundary-setting clusters; mindful unplugging prompts; screen-time pacing

Balanced Traditionalist → Simplified UI with larger text; face-to-face hybrid support; guided step-by-step onboarding

Tech-Fluent → Advanced self-monitoring dashboards; peer mentoring features; AI-personalized content feeds

Disconnected → Guided onboarding with tech literacy scaffolding; low-barrier SMS/voice engagement; community navigator support

Each phenotype receives a tailored AI pathway — moving digital mental health from uniform delivery to precision care.

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